

HANANEEL EPHRAIM MANDAM

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EDUCATION

B.E. Mechanical Engineering | BITS Pilani, Hyderabad Campus

Expected 2028

Coursework: Artificial Intelligence, Vibrations & Control, Mechanisms & Machines, Advanced Mechanics of Solids, Fluid Mechanics, Principles of Aerodynamics, Heat Transfer, Differential Equations

EXPERIENCE

Research Intern, Physiological Signal Processing | CSIR-CEERI, Chennai

May 2026 – Jul 2026

- Built an end-to-end real-time ECG/HRV pipeline that ingests commercial health-band data and estimates KSS drowsiness levels over live rolling windows, for a defense project.
- Improved Pan-Tompkins++ R-peak detection to **98.8%** accuracy on the MIT-BIH NSTDB benchmark (24 dB and 18 dB SNR), vs. **81.7%** for NeuroKit2.
- Designed a pub/sub ECGBus architecture for real-time multi-subject BLE session management, with hardware-level signal quality gating on Polar H10 straps and automated logging of vehicle and software crash events.

Powertrain Engineer & Electrical Lead | Electric Baja SAE, BITS Pilani Hyderabad

Apr 2025 – Apr 2026

- Designed a 4:1 inline spur gear reduction and chain-sprocket drivetrain in SolidWorks; validated shaft and gear geometry under **90 Nm** peak torque in ANSYS FEA.
- Engineered the full vehicle safety interlock system on microcontrollers to SAE E-Baja rules; designed and implemented the complete wiring schematic.
- Retrofitted chassis members via profiling and welding for the electric drivetrain conversion; coordinated a **7-person** team through fabrication and vendor timelines.

PROJECTS

Uncertainty-Aware UAV Emergency Landing Zone Selection | [GitHub](#)

Jun 2026 – Present

- Built a three-channel risk fusion pipeline: semantic segmentation (SegFormer-B2 fine-tuned on UAVid), epistemic uncertainty via MC Dropout, and temporal label-disagreement tracking.
- Ablations on UAVid and Aeroscapes: temporal consistency alone still marked unsafe zones as safe on both datasets; every uncertainty-aware variant produced **zero** false-safe selections.

Ground-Effect Wing CFD | Principles of Aerodynamics Coursework

In Progress

- Built a 2D RANS workflow (k-omega SST) in ANSYS Fluent for an inverted NACA 6412 wing; swept angle of attack from -12° to $+4^\circ$ to map downforce, drag, and stall.
- Linear-range lift slope of **0.101/deg** and zero-lift angle within **0.3°** of thin-airfoil theory; peak L/D of **62**. Next: ride-height study above a ground plane.

3D-Printed Sarrus Linkage | Mechanisms & Machines Coursework

Feb 2026

- Designed, 3D printed, and assembled a functional Sarrus linkage; validated constrained straight-line motion analytically and experimentally.

SKILLS

Programming: Python (PyTorch, NumPy, OpenCV, HuggingFace), MATLAB, Simulink, C, Java

CAD, FEA & CFD: SolidWorks, ANSYS (Static Structural, Modal, Fluent)

ML & Vision: Semantic Segmentation, Uncertainty Quantification, Transfer Learning

Fabrication: MIG/TIG Welding, Lathe, Sheet Metal, FDM 3D Printing, Soldering

Tools: Git, Linux, Weights & Biases